ERSATZTEILSTORE · APPLICATION TASK

How fixpart.de structures 20 million spare parts

Category analysis, comparison with our three API shops, and a practical rebuild plan for Laravel, PostgreSQL and Meilisearch.

PREPARED FOR	DATE	SCOPE
Ersatzteilstore	November 2026	fixpart.de + 3 API shops

THE SHORT ANSWER

FixPart did not build a bigger tree. It built a product database with a curated navigation on top.

105
appliance groups
(Waschmaschine, Backofen …)

1,511
brands in the A–Z directory
(site claims 2,000+)

375
browsable part categories
plus a 5-level classification

3
navigable levels + the
type-number filter

> **Why it feels so much better than a wholesaler shop:** the API supplies the facts — article numbers, brands, device types, compatibility. FixPart adds the editorial layer: its own appliance names, part categories, classification, SEO landing pages and fit guarantee. **That layer is the product.**

WHAT WE ANALYSED

105 appliance group pages, 1,511 brand pages, 375 part category pages, 39 product pages with their internal classification, plus the category tree of ersatzteilblitz.de, wildbihler-shop.de and ersatzteile-24.com.

WHAT WE DELIVER

A 2-page result, the full category tree as CSV/JSON, and the implementation code (migration, importer, Meilisearch indexer) as a separate archive.

THE CORE FINDING

Two axes, one catalogue — not one deep category tree

A customer either starts with **their appliance** or with **the part they need**. FixPart serves both from the same products, without duplicating them.



Consequence for our rebuild: the category tree is a view, not the data model. The data model is a product catalogue with two dimensions and a compatibility table.

THE STRUCTURE IN NUMBERS

What the tree actually contains, level by level

AXIS	LEVEL	NODES	EXAMPLE
Device	**L1 · Appliance group**	**105**	Waschmaschine, Kühlschrank, Backofen
Device	**L2 · Brand**	**1,511**	Bosch, Siemens, Miele
Device	**L3 · Model / type number**	**thousands / brand**	WGG244Z40 — only reachable through the type-number filter
Part	**L1 · Browsable part category**	**375**	Türgummi, Türfach, Pumpe, Kohlefilter
Part	**L2–L5 · Internal classification**	**5 levels**	Mechanisch › Gehäuseteil › Fenster › Tür

What the 5 classification levels contain

LEVEL	VERIFIED NODES*	EXAMPLE
L1	**3**	Mechanisch
L2	**7**	Gehäuseteil
L3	**14**	Fenster
L4	**8**	Tür
L5	**1**	Türfach

Important nuance

FixPart markets "5,000+ categories", but only **375** have a public index page. The rest are classification leaves and generated appliance × brand × part combinations. The classification is visible on the product page, not as a clickable sub-tree.

* Verified sample: 33 nodes / 39 product pages. The full classification only exists in the supplier feed, so these counts are evidence, not a complete census.

HOW THE CUSTOMER MOVES

Device-first navigation, built from a fixed URL matrix

Every combination of appliance, brand and part gets its own landing page. Breadcrumbs come from the URL, not from the classification.

1

Choose the appliance

`/gerate`

105 appliance groups. This is the main entrance and the strongest SEO layer.

2

Choose the brand

`/waschmaschine-ersatzteile`

Brand tiles, part-category tiles and popular products on the appliance page.

3

Narrow to the part

`/waschmaschine-ersatzteile/bosch`

Product list plus the type-number picker; filters for brand and appliance.

4

Deepest landing page

`/waschmaschine-ersatzteile/bosch/turgummi`

Appliance × brand × part. This is the page that ranks and converts.

THE SECOND ENTRANCE

People who know the part name start at **/ersatzteile/turgummi**. The 375 part categories are a flat A–Z index — siblings, not a hierarchy. All other combinations are generated from the same two axes.

BREADCRUMB · PART PAGE

Home › Waschmaschine Ersatzteile › Türgummi

BREADCRUMB · PRODUCT

Home › Tür › Product — only the leaf is shown

EFFECTIVE FILTERS

DIMENSIONS

Appliance type · brand · model / type number · part category

LISTING

Price · availability · condition (original / aftermarket / refurbished)

THE FULL URL MATRIX

`/{appliance}-ersatzteile` · `/{appliance}-ersatzteile/{brand}` · `/{appliance}-ersatzteile/{brand}/{part}` · `/ersatzteile/{brand}/{part}` · `/produkt/{slug}`

THE CLEVER PART

One product, one category — but dozens of listings

The sample product **Bosch 00704286** is a washing-machine door. It has exactly **one** classification path, yet it is reachable from every compatible appliance and brand.

CLASSIFICATION

Mechanisch › Gehäuseteil › Fenster › Tür

DEVICE BRANDS

Balay · Bosch · Profilo · Siemens

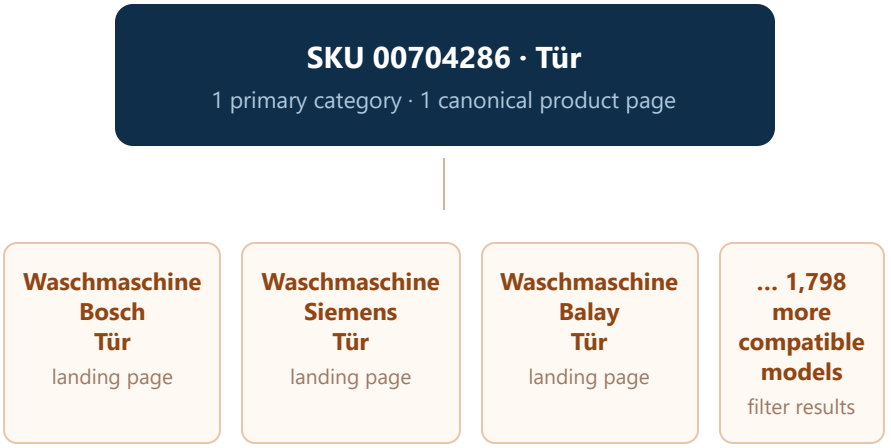
APPLIANCE TYPE

Waschmaschine

COMPATIBLE DEVICES

1,802 models

No category explosion. The product is not copied into 1,802 categories. Compatibility links make it appear under the right appliance × brand × part pages, while the product page stays canonical.



Same SKU, many entry points — one stock record, one price, one product page.

No duplicate content
One canonical product URL, however many listings link to it.

One source of truth
Price and stock are updated once, not per category.

SEO at scale
Thousands of relevant landing pages from one catalogue.

COMPARISON

Same wholesaler data — very different shop

The three API shops publish the supplier's raw dimensions. FixPart normalises them and adds a curated layer on top.

SHOP	NAVIGATION	DEPTH	OWN TAXONOMY / EDITORIAL LAYER
ersatzteilblitz.de	Gerätegruppen A–Z + Artikelgruppen A–Z; inquiries deep-link to shop.euras.com	2	none supplier dimensions only
wildbihler-shop.de	Hersteller A–Z + Geräte A–Z; brand × device treated as separate "manufacturers" (e.g. AEG-DAMPFBÜGELEISEN)	2–3	none brand and device stay raw
ersatzteile-24.com	Brand index, brand → device model lists, part pages; a repair assistant at /shop.php	3	none no renamed groups or part taxonomy
fixpart.de	105 appliance groups → 1,511 brands → 375 part categories → type number; separate A–Z part index	3 + 5	yes renamed groups, part categories, classification, SEO landings, fit guarantee

Identical in all four: part brand, device brand, device type, model / type number, article number, description and compatibility all come from the same supplier dimensions.

The difference is editorial: FixPart turns dimensions into a shop. The others stay databases — even when the underlying articles are the same.

Our starting point: we already have the same dimensions. What we lack is the 375-node part taxonomy and the generated landing pages — **both are buildable on top of the feed.**

WHAT WE GET, WHAT WE BUILD

The API gives us the facts. The taxonomy is our work.

PROVIDED DIRECTLY BY THE WHOLESALER API

- ✓ Article number, MPN and description
- ✓ Supplier part group / part type in supplier wording
- ✓ Device brand(s), device type(s), model / type numbers
- ✓ Compatibility list (the most valuable field)
- ✓ Price and stock

WHAT WE MUST ADD (THE MOAT)

- + 105 normalised appliance groups (one name per device type)
- + 375 part categories + 5-level classification
- + Aliases and synonyms (supplier spelling → our category)
- + SEO copy, landing pages, breadcrumbs, sitemaps
- + Fit guarantee, translations for DE / ES / IT

> **Avoid the trap of the API shops:** never model “AEG-Waschmaschine” as a category. Brand and device are separate dimensions, connected to products through compatibility.

Budget the taxonomy
The import is commodity work. The curated categories and aliases are the differentiator.

Curate once, reuse
One taxonomy serves DE, ES and IT; only labels and copy need translating.

The SEO follows
Once the two axes are clean, landing pages are generated, not hand-built.

THE REBUILD

One import pipeline, two databases, one storefront

PostgreSQL owns the hierarchy and the truth. Meilisearch owns search, facets and the type-number finder.

SOURCE
Wholesaler API
Articles, brands, device types, compatibility, price, stock

›

STEP 1
Staging
Raw JSON + hash, so every re-import is idempotent and debuggable

›

STEP 2
Normaliser
Supplier spelling → our brand, appliance and part category

›

STEP 3
PostgreSQL
categories · brands · models · products · compatibility · SEO landings

›

STEP 4
Meilisearch
One product index with facets; separate model index

›

STEP 5
Storefront
Category pages, search, breadcrumbs, type-number finder

Category table: adjacency list (parent_id) plus a materialised dot path. A subtree is one indexed *LIKE 'mechanisch.gehaeuseteil.%'* — no recursive query per request.

Search index: flat *category_ids*, *brand*, *appliance_types*, *condition*, *price*, *in_stock* as filterable facets. The tree and breadcrumbs stay in Postgres.

Scale: ~500 category rows (105 appliance + 375 part + intermediate levels) — small enough to cache in Redis. The compatibility table can reach millions of rows.

Rule: a product gets one primary category from the feed. Appliance and brand listings are driven by compatibility, never by extra category assignments.

DELIVERY PLAN

Start with the catalogue, then add the SEO surface

WEEKS 1–3

Taxonomy + import

105 appliances, 375 part categories, aliases, nightly import, product & compatibility tables, canonical product pages.

WEEKS 4–6

Navigation + search

Appliance → brand → part pages, Meilisearch facets, type-number finder, breadcrumbs, filters.

WEEKS 7–10

SEO landings

Generated appliance × brand × part pages, sitemaps, internal linking. Publish only pages with ≥ 3 products, otherwise noindex.

ONGOING

Expansion

DE → ES / IT translations, synonym tuning, zero-result monitoring, price/stock freshness.

Indicative effort for one senior developer; actual timing depends on the feed contract and catalogue size.

DECISIONS WE NEED FROM THE BUSINESS

- Wholesaler contract and field list — does the feed already contain a product-group path?
- If not: who owns the 375-node taxonomy and the 5-level classification?
- Which market launches first — and do we translate categories or only products?
- Minimum assortment per landing page before it is indexable?

ASSUMPTIONS

EURAS-type feed (ersatzteilblitz deep-links to shop.euras.com). Brand count 1,511 captured vs 2,000+ claimed. “5,000+ categories” = classification leaves + generated combinations, not 5,000 index pages. Build from our own feed and editorial work — not scraped FixPart content.

WHAT WE DELIVER

A clear model, a working plan, and the code to start Monday.

The analysis is evidence-based: public pages were captured, parsed and counted. Every number in this deck is reproducible from the data files.

01 · PRESENTATION

This deck

The strategy and the findings for a non-technical audience.

02 · FINAL RESULT

2-page report + tree

Executive summary, numbers, comparison, open questions. CSV/JSON tree included.

03 · CODE

Working implementation

Laravel migration, importer, Meilisearch indexer, settings and full research data.

NEXT STEP · 01

Confirm the wholesaler feed fields with the API team.

NEXT STEP · 02

Agree who owns the category taxonomy.

NEXT STEP · 03

Pick the first launch market: DE, ES or IT.